Assignment 4

Packet Tracer - TCP and UDP Communications

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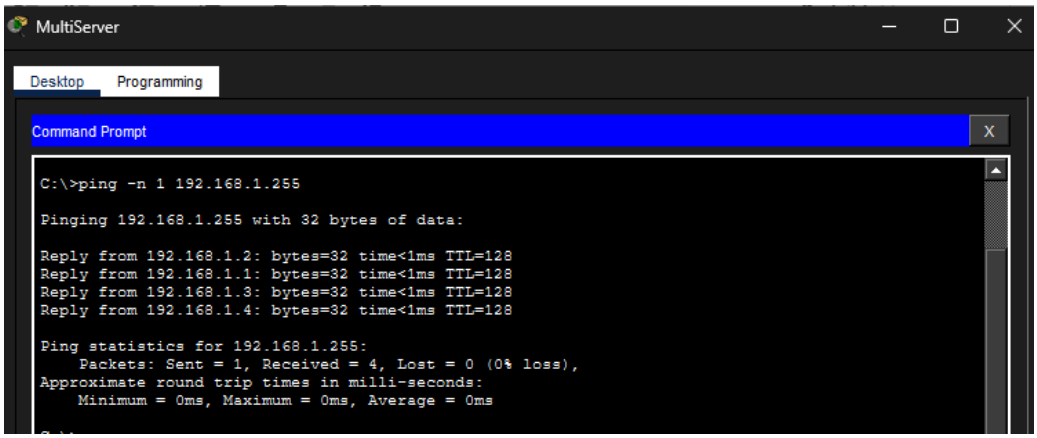
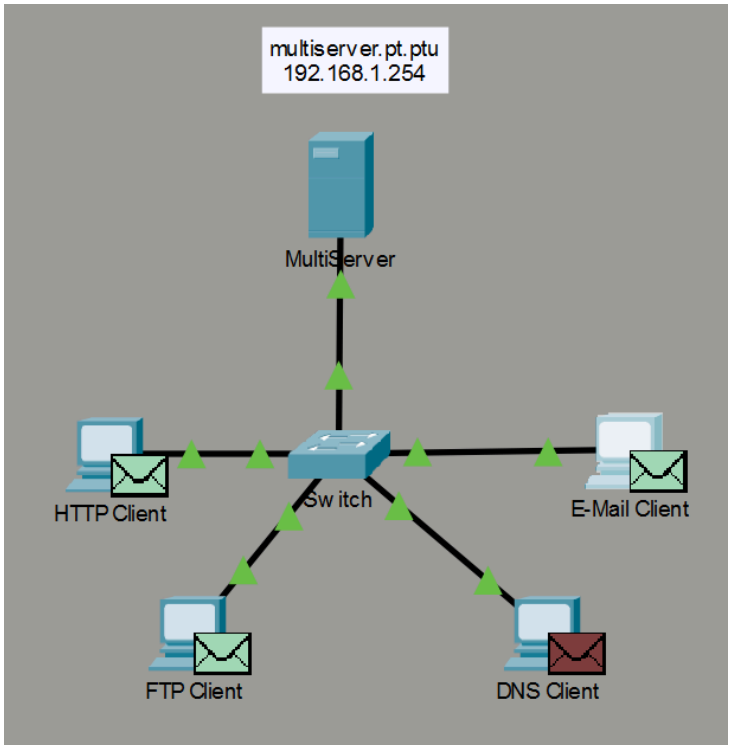
PART1

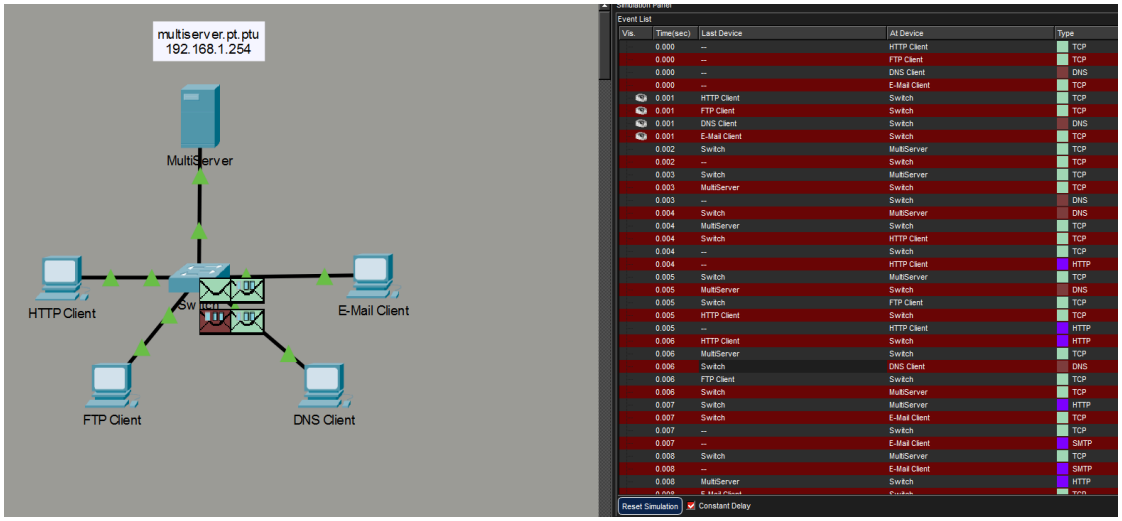
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

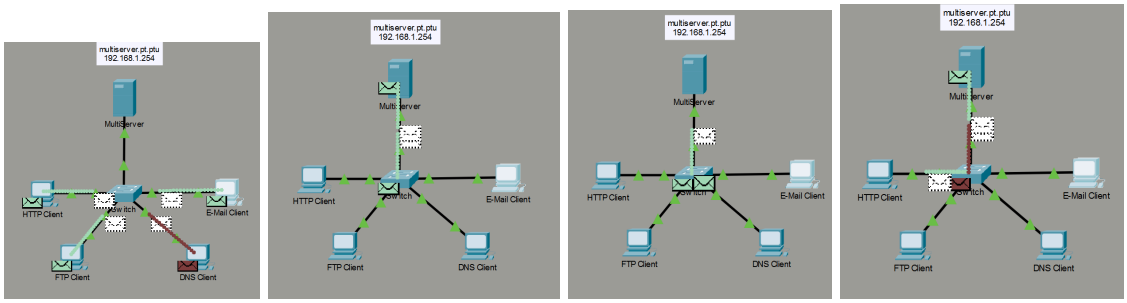


Step 2-5: Generate web (HTTP) traffic , FTP traffic ,DNS traffic And Email traffic.
Step 6: Verify that the traffic is generated and ready for simulation.



Step 7: Examine multiplexing as the traffic crosses the network.





Question:

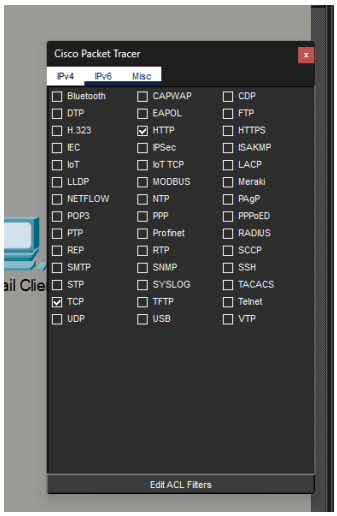
What is this called?
=Multiplexing

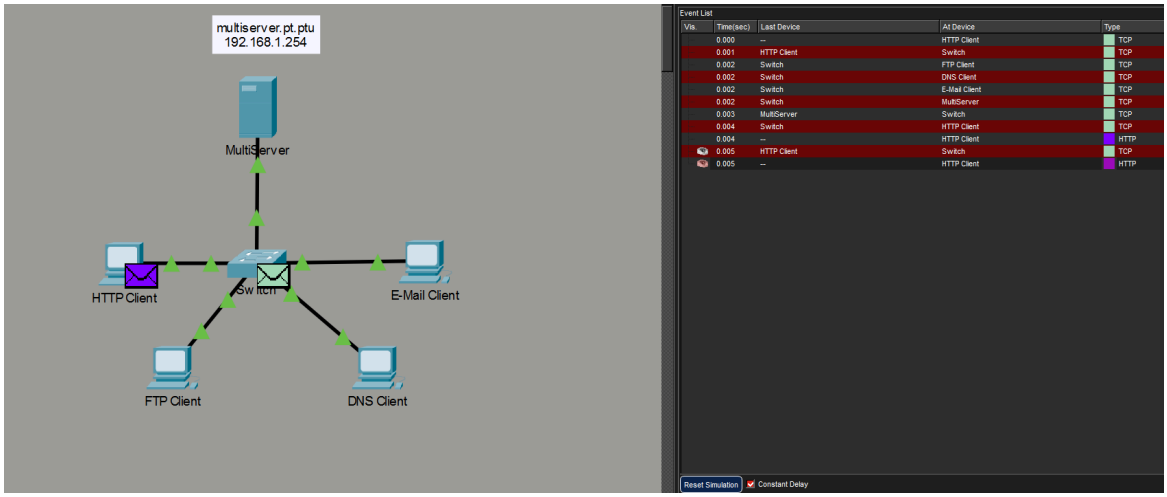
A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?
=แต่ละสีแทนแต่ละโปรโตคอล (HTTP, TCP, DNS, UDP ฯลฯ)

PART 2

Step 1: Examine HTTP traffic as the clients communicate with the server.

1.1.Reset Simulation and Filter For HTTP and TCP

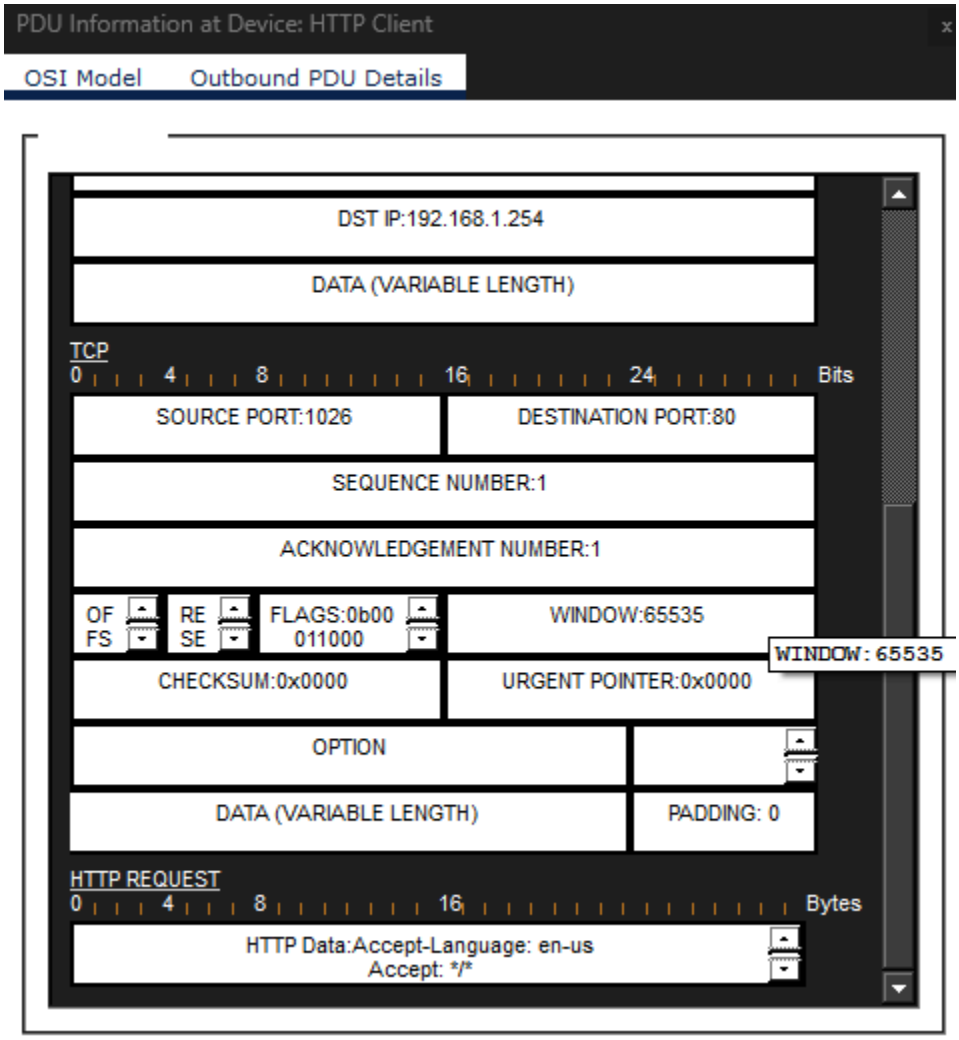




Question:

Why did it take so long for the HTTP PDU to appear?
=TCP ต้องทำ 3-Way Handshake (SYN → SYN-ACK → ACK) ก่อน

1.2. Click the PDU envelope to show the PDU details. Click the Outbound PDU Details tab and scroll down to the second to the last section.



Question:

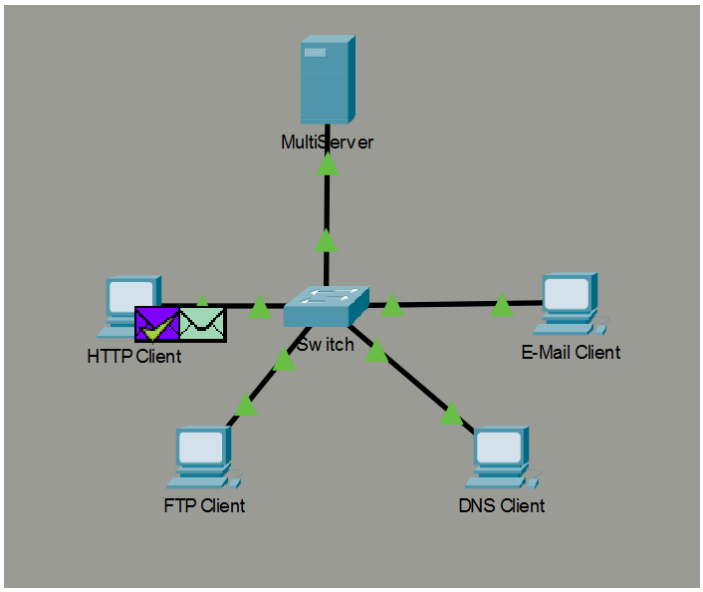
What is the section labeled?
=TCP

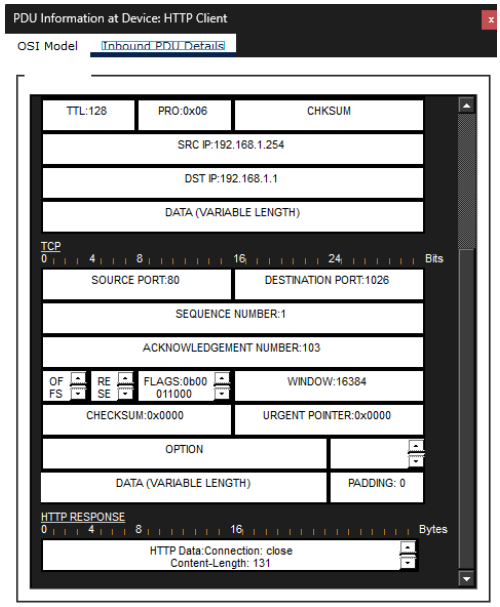
Are these communications considered to be reliable?
=Yes TCP มีกลไกการตรวจสอบความถูกต้องในการการตอบรับข้อมูล

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values.
= SRC PORT : 1026
DEST PORT : 80
SEQUENCE NUM : 1
ACK NUM : 1

Which TCP flags are set in this PDU?
= ACK and PSH

1.3 Close the PDU and click Capture/Forward until a PDU with a checkmark returns to the HTTP Client.





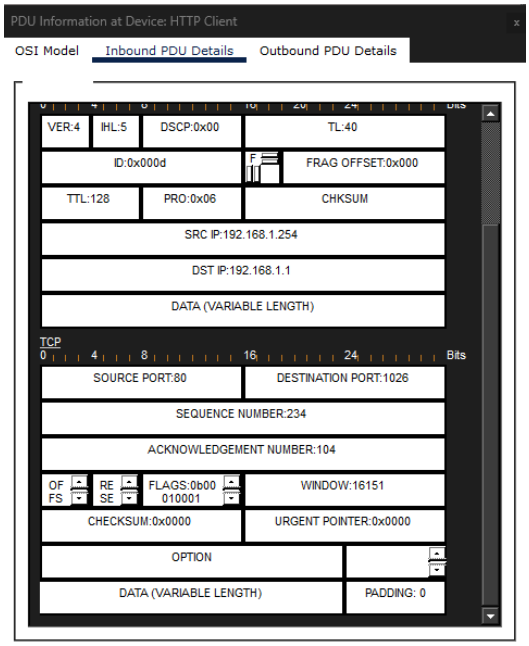
Question:

How are the port and sequence numbers different than before?
=

Before
SRC PORT : 1026
DEST PORT : 80
SEQUENCE NUM : 1
ACK NUM : 1

After
SRC PORT : 80
DEST PORT : 1026
SEQUENCE NUM : 1
ACK NUM : 103

1.4.Click the HTTP PDU which HTTP Client has prepared to send to MultiServer. This is the beginning of the HTTP communication. Click this second PDU envelope and select Outbound PDU Details.



Question:

What information is now listed in the TCP section?

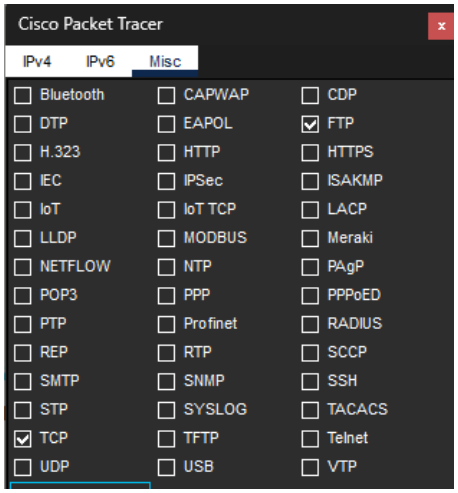
= SRC PORT : 80
DEST PORT : 1026
SEQUENCE NUM : 234
ACK NUM : 104

How are the port and sequence numbers different from the previous two PDUs?

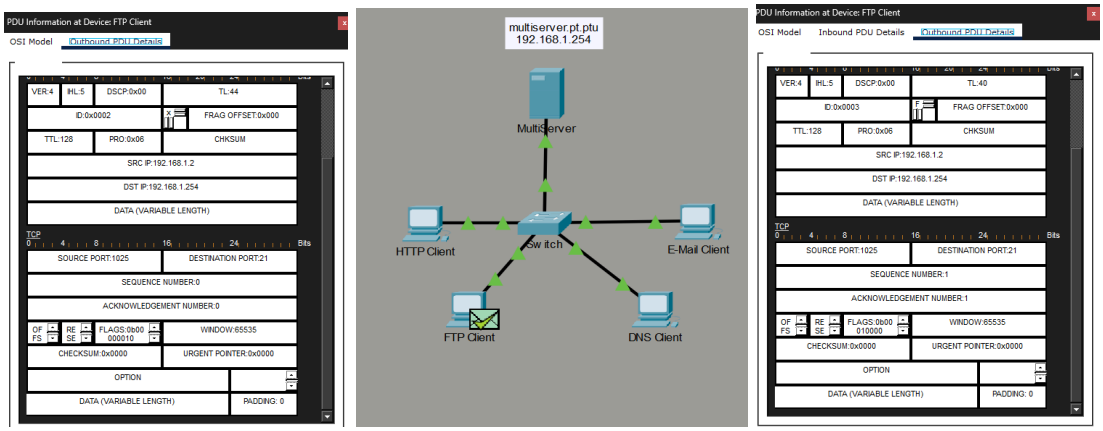
= Source port กับ Destination port สลับกัน Sequence NUM กับ ACK NUM เพิ่มขึ้น

Step 2: Examine FTP traffic as the clients communicate with the server.

2.1.Reset Simulation and Filter For FTP and TCP



2.2 Observe TCP



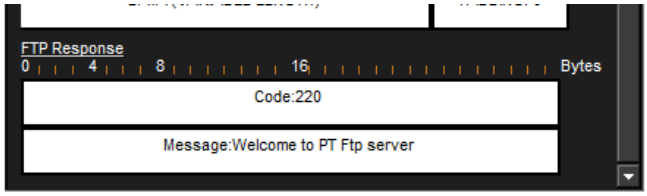
Question:

Are these communications considered to be reliable?
= Yes

What is the value in the flag field?
= 0b00000010 (SYN)

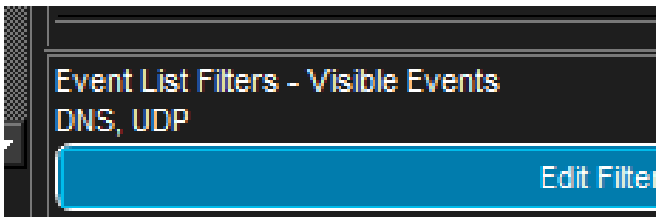
How are the port and sequence numbers different from the previous results?
= portสลับกัน numberจาก0เป็น1

What is the message from the server?
=Welcome to PT Ftp server

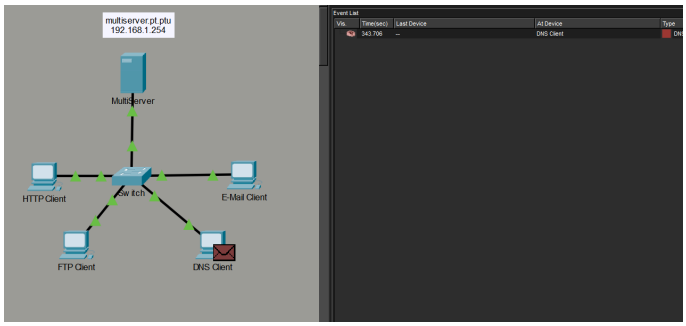


Step 3: Examine DNS traffic as the clients communicate with the server.

3.1 Reset Simulation and filter only DNS and UDP



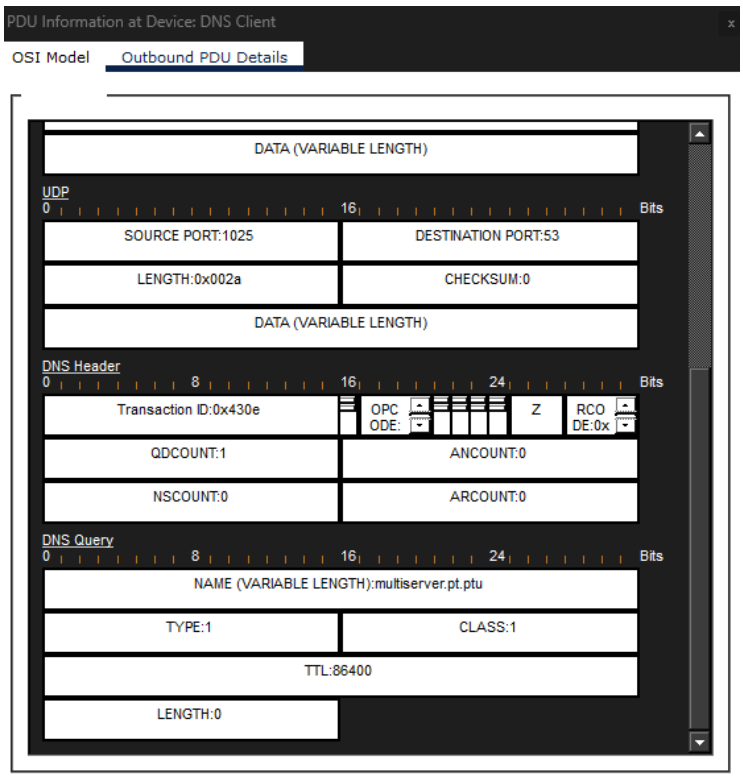
3.2 Observe



Question:

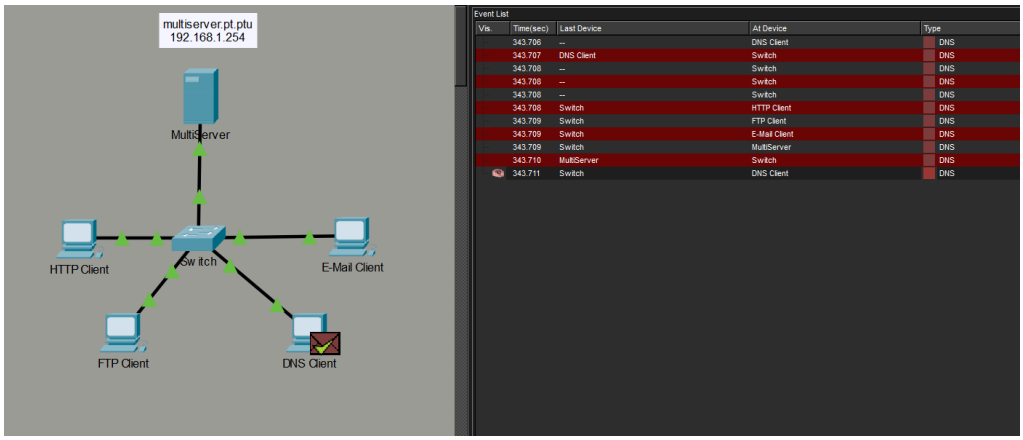
What is the Layer 4 protocol?
= UDP

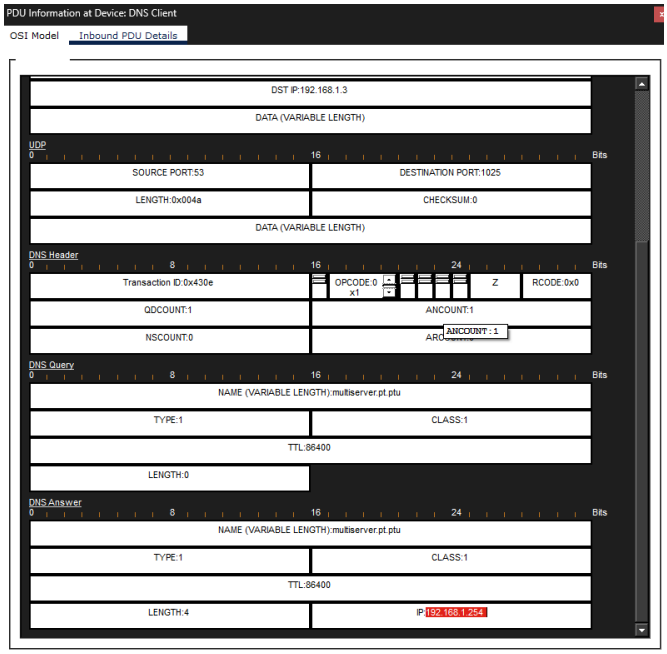
Are these communications considered to be reliable?
= No ไม่มีธงไม่มีลำดับ



Question:

Why are there no sequence and acknowledgement numbers?
= เพราะเป็น UDP เน้นเร็วไม่ได้ตรวจลำดับข้อมูล





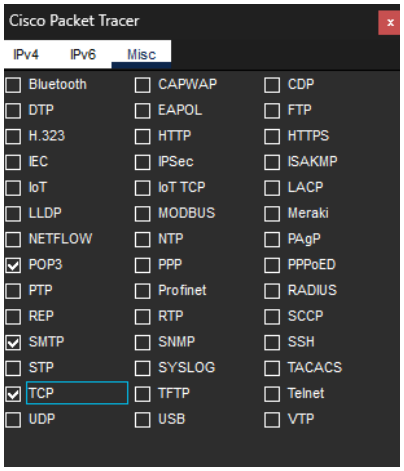
Question:

How are the port and sequence numbers different than before?
= Source กับ Destination สลับกัน

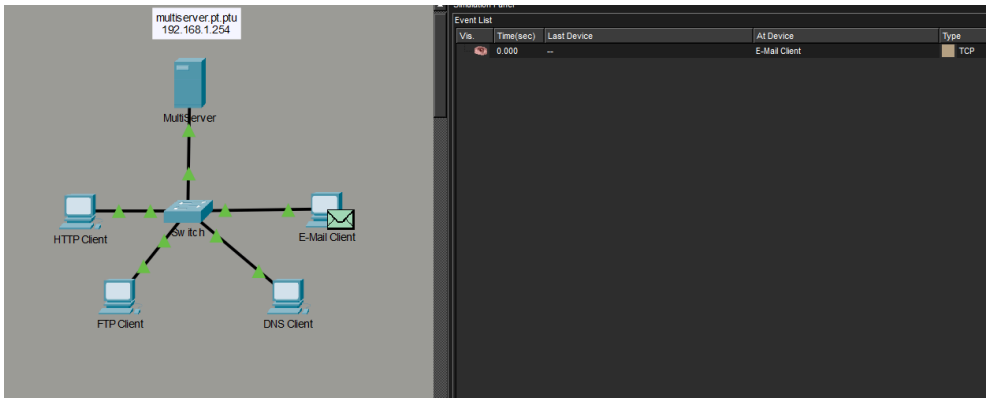
What is the last section of the PDU called? What is the IP address for the name multiserver.pt.ptu?
=192.168.1.254

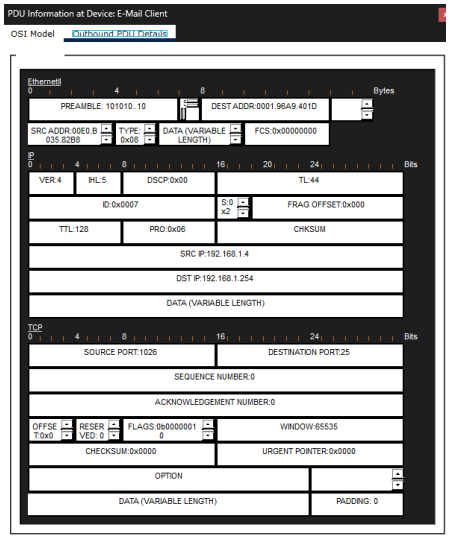
Step 4: Examine email traffic as the clients communicate with the server.

4.1 Reset Simulation and filter only POP3 , SMTP and TCP



4.2 Observe the PDU





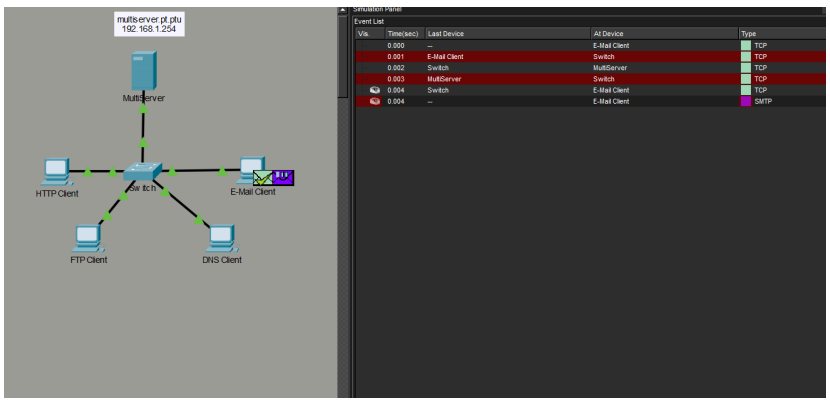
Question:

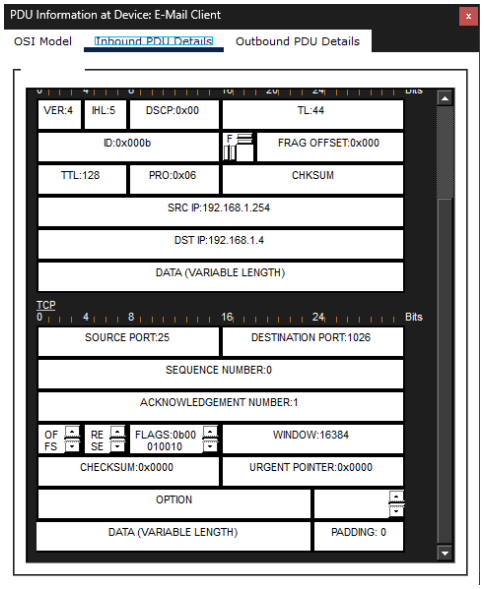
What transport layer protocol does email traffic use?
= TCP

Are these communications considered to be reliable?
= YES

Record the SRC PORT, DEST PORT, SEQUENCE NUM, and ACK NUM values. What is the flag field value?

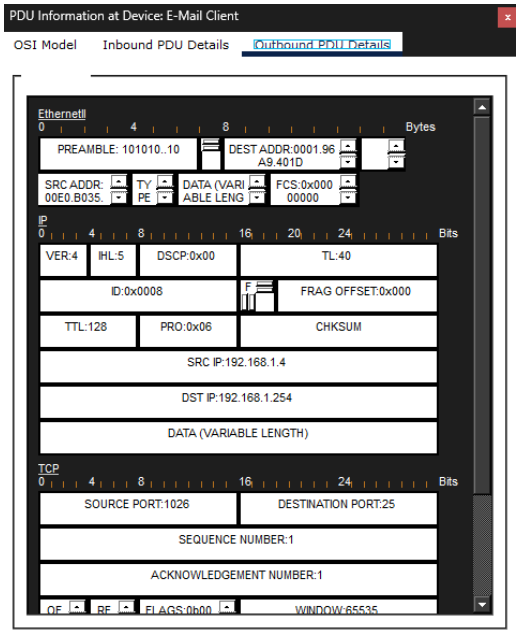
= SRC PORT : 1026
DEST PORT : 25
SEQUENCE NUM : 0
ACK NUM : 0
FLAG : 0b00000010





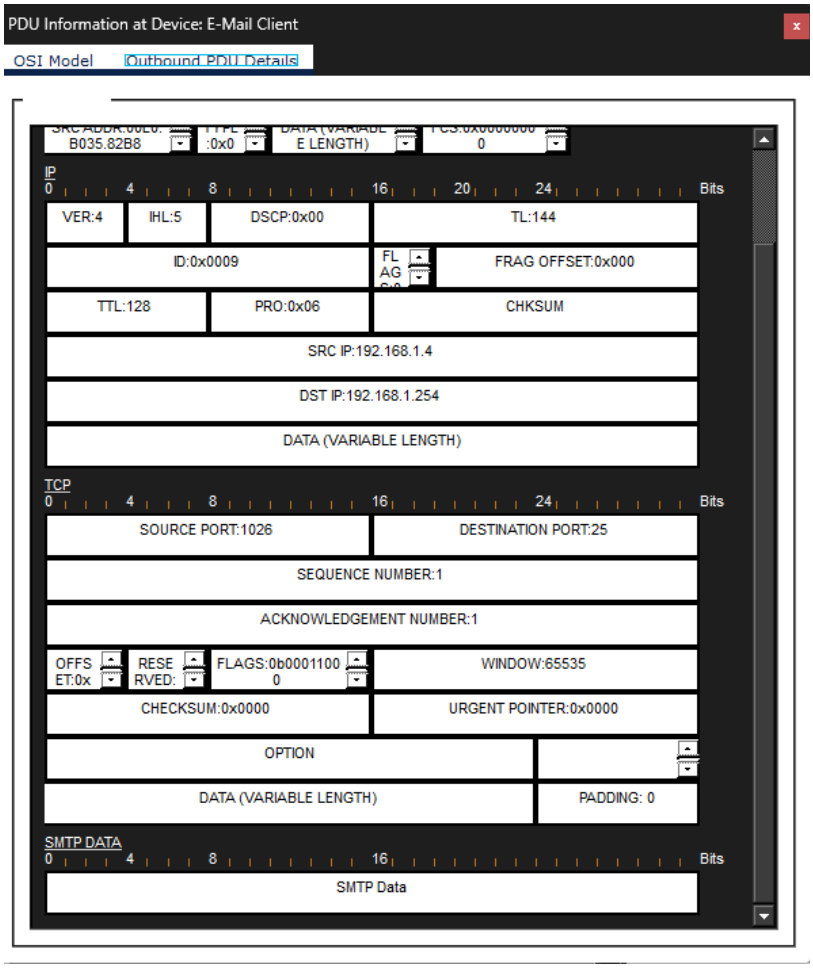
Question:

How are the port and sequence numbers different than before?
= portสลับกัน และ acknowledge num จาก 0 เป็น1



Question:

How are the port and sequence numbers different from the previous two results?
=Port เหมือนตอนเริ่ม sequence num จาก0 เป็น 1



Question:

How are the port and sequence numbers different from the previous two **PDU**s?
= port และ sequence numbers เหมือน outbound อันก่อนหน้า
ส่วนเทียบกับ inbound port จะสลับกัน และ sequence number ต่างกัน (0กับ1)

What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?
= PORT 25 SMTP
PORT 110 POP3